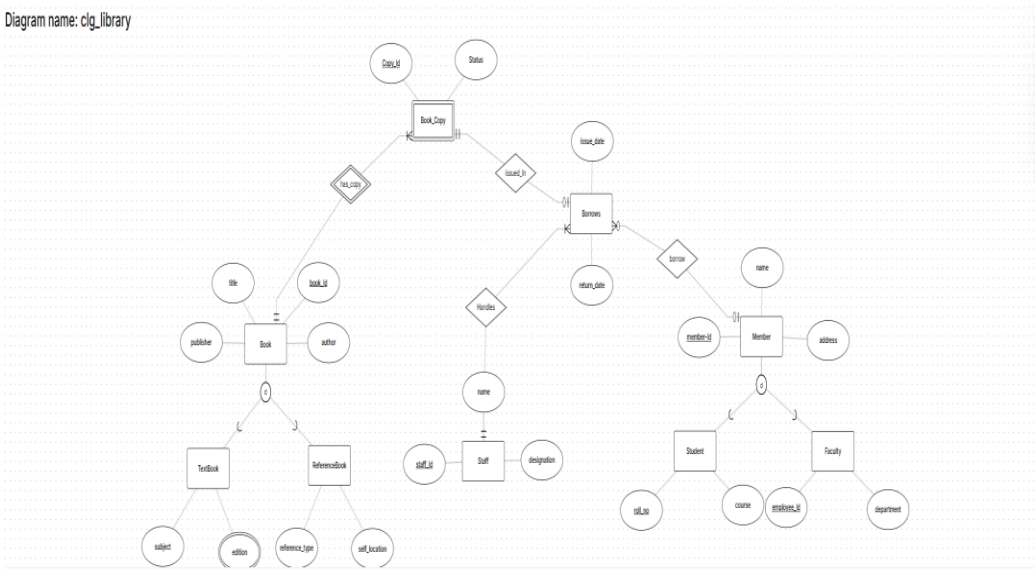
DBMS ALL Final Codes

(Daredevil)

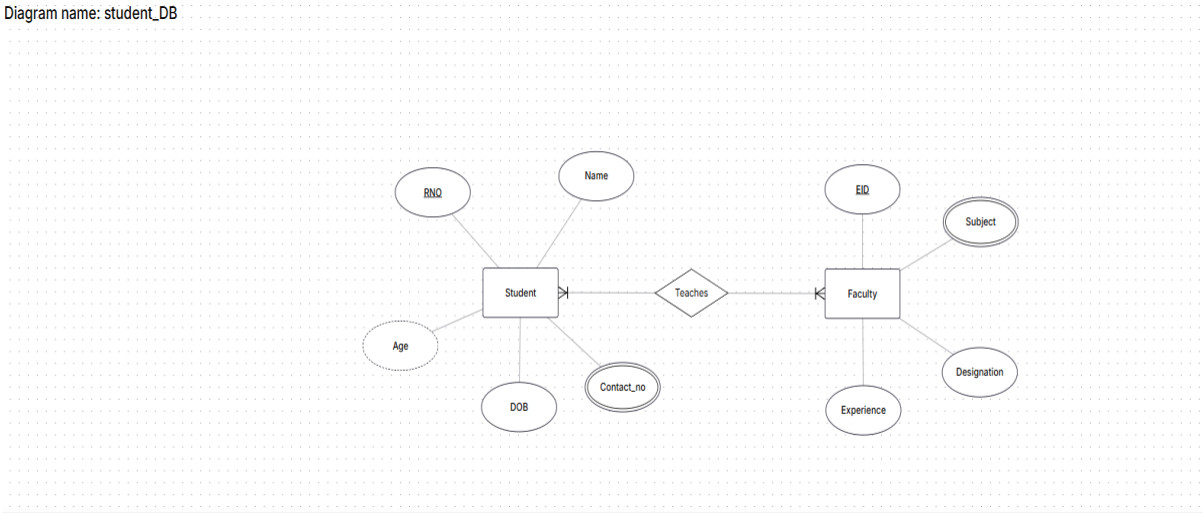
1) E.R Diagram....

Library



rtisement

Student



rtisement

2) DDL/DML

```
SQL> CREATE TABLE customer (
  2     cust_id NUMBER(5) PRIMARY KEY,
  3     cust_name VARCHAR2(20) NOT NULL,
  4     email VARCHAR2(30) UNIQUE,
  5     phone NUMBER(10) UNIQUE,
  6     city VARCHAR2(15) DEFAULT 'Pune'
  7  );

Table created.

SQL> CREATE TABLE account (
  2     acc_no NUMBER(10) PRIMARY KEY,
  3     cust_id NUMBER(5),
  4     acc_type VARCHAR2(10) NOT NULL,
  5     balance NUMBER(10,2) DEFAULT 0,
  6     CONSTRAINT fk_cust FOREIGN KEY (cust_id)
  7     REFERENCES customer(cust_id)
  8  );

Table created.

SQL> CREATE TABLE transaction (
       txn_id NUMBER(10) PRIMARY KEY,
     acc_no NUMBER(10),
  amount NUMBER(10,2) NOT NULL,
      txn_type VARCHAR2(10),
    txn_date DATE DEFAULT SYSDATE,
    CONSTRAINT fk_acc FOREIGN KEY (acc_no)
       REFERENCES account(acc_no)
   );

Table created.
```

DML

```
SQL> INSERT INTO customer
  VALUES (&cust_id, '&cust_name', '&email', &phone, '&city');
Enter value for cust_id: 101
Enter value for cust_name: Anmol
Enter value for email: anmol@gmail.com
Enter value for phone: 754985
Enter value for city: jammu
old   2: VALUES (&cust_id, '&cust_name', '&email', &phone, '&city')
new   2: VALUES (101, 'Anmol', 'anmol@gmail.com', 754985, 'jammu')

1 row created.

SQL> /
Enter value for cust_id: 102
Enter value for cust_name: Arnav
Enter value for email: jjfjdfhj
Enter value for phone: 89786
Enter value for city: Pune
old   2: VALUES (&cust_id, '&cust_name', '&email', &phone, '&city')
new   2: VALUES (102, 'Arnav', 'jjfjdfhj', 89786, 'Pune')

1 row created.

SQL> INSERT INTO account
  2  VALUES (&acc_no, &cust_id, '&acc_type', &balance);
Enter value for acc_no: 1001
Enter value for cust_id: 101
Enter value for acc_type: saving
Enter value for balance: 5000
old   2: VALUES (&acc_no, &cust_id, '&acc_type', &balance)
new   2: VALUES (1001, 101, 'saving', 5000)

1 row created.
```

```
SQL> /
Enter value for acc_no: 1002
Enter value for cust_id: 102
Enter value for acc_type: current
Enter value for balance: 10000
old   2: VALUES (&acc_no, &cust_id, '&acc_type', &balance)
new   2: VALUES (1002, 102, 'current', 10000)

1 row created.

SQL> INSERT INTO transaction
  2  VALUES (&txn_id, &acc_no, &amount, '&txn_type', SYSDATE);
Enter value for txn_id: 1
Enter value for acc_no: 1001
Enter value for amount: 2000
Enter value for txn_type: deposit
old   2: VALUES (&txn_id, &acc_no, &amount, '&txn_type', SYSDATE)
new   2: VALUES (1, 1001, 2000, 'deposit', SYSDATE)

1 row created.

SQL> /
Enter value for txn_id: 2
Enter value for acc_no: 1002
Enter value for amount: 5000
Enter value for txn_type: withdraw
old   2: VALUES (&txn_id, &acc_no, &amount, '&txn_type', SYSDATE)
new   2: VALUES (2, 1002, 5000, 'withdraw', SYSDATE)

1 row created.

SQL> SELECT * FROM customer;

   CUST_ID CUST_NAME          EMAIL                          PHONE
```

---------- ---------- ---------- ----------
CITY
---
101 Anmol anmol@gmail.com 754985
jammu

102 Arnav jjfjdfhj 89786
Pune

SQL> SELECT * FROM account;

ACC_NO CUST_ID ACC_TYPE BALANCE
---------- ---------- ---------- ----------
1001 101 saving 5000
1002 102 current 10000

SQL> SELECT * FROM transaction;

TXN_ID ACC_NO AMOUNT TXN_TYPE TXN_DATE
---------- ---------- ---------- ---------- ---------
1 1001 2000 deposit 17-APR-26
2 1002 5000 withdraw 17-APR-26

SQL> UPDATE account SET balance = balance + 1000 WHERE acc_no = 1001;

1 row updated.

SQL> DELETE FROM transaction WHERE txn_id = 2;

1 row deleted.

DCL

CREATE USER bankuser IDENTIFIED BY 1234;
GRANT SELECT, INSERT, UPDATE ON customer TO bankuser;
REVOKE UPDATE ON customer FROM bankuser;

3) Sequence, View, Index, Synonym

SQL> create table custanm17(cust_id number PRIMARY KEY, name varchar(25),balance number(9),branch varchar(20));
Table created.
SQL> create sequence cust_seq start with 101 increment by 1;
Sequence created.
SQL> insert into custanm17 values(cust_seq.nextval,'&name',&balance,'&branch');
Enter value for name: anmol Enter
value for balance: 80000 Enter
value for branch: 1
old 1: insert into custanm17 values(cust_seq.nextval,'&name',&balance,'&branch') new
1: insert into custanm17 values(cust_seq.nextval,'anmol',80000,'1')
1 row created.

SQL> create table accanm17(acc_id number,trans_id number(7),name varchar(30),balance number(9));
Table created.
SQL> create sequence trans_seq start with 501 increment by 1;
Sequence created.
SQL> create sequence acc_seq start with 101 increment by 1;

Sequence created.
SQL> insert into accanm17 values(acc_seq.nextval,trans_seq.nextval,'shradha',50000);
1 row created.
SQL> desc accanm17
Name Null? Type
--- --- ---
ACC_ID NUMBER
TRANS_ID NUMBER(7)
NAME VARCHAR2(30)
BALANCE NUMBER(9)
SQL> select * from custanm17;
CUST_ID NAME BALANCE BRANCH
--- --- --- ---
101 anmol 80000 1
SQL> select * from accanm17;
ACC_ID TRANS_ID NAME BALANCE
--- --- --- ---

101 501 shradha 50000
SQL> insert into accanm17
values(acc_seq.nextval,trans_seq.nextval,'rahul',
40000);
1 row created.
SQL> select * from accanm17;
ACC_ID TRANS_ID NAME BALANCE
--- --- --- ---
101 501 shradha 50000 102 502
rahul 40000
SQL> insert into accanm17
values(acc_seq.nextval,trans_seq.nextval,'aditya'
,60000);
1 row created.
SQL> select * from accanm17;
ACC_ID TRANS_ID NAME BALANCE
--- --- --- ---

101 501 shradha 50000 102 502
rahul 40000
103 503 aditya 60000
SQL> insert into custanm17
values(cust_seq.nextval,'&name',&bal
ance,'&branch');
Enter value for name: jyoti
Enter value for balance: 45000 Enter
value for branch: 5
old 1: insert into custanm17
values(cust_seq.nextval,'&name',&bal
ance,'&branch') new

1: insert into custanm17
values(cust_seq.nextval,'jyoti',45000,'5')
1 row created.
SQL> select * from custanm17;
CUST_ID NAME BALANCE BRANCH
---------- ---------- ---------- ----------
101 anmol 80000 1
102 jyoti 45000 5
SQL> create synonym acc for accanm17;
Synonym created.
SQL> select * from accanm17;
ACC_ID TRANS_ID NAME BALANCE
---------- ---------- ---------- ----------
101 501 shradha 50000 102 502
rahul 40000
103 503 aditya 60000
SQL> set timing on;
SQL> select * from accanm17;
ACC_ID TRANS_ID NAME BALANCE
---------- ---------- ---------- ----------
101 501 shradha 50000 102 502
rahul 40000
103 503 aditya 60000
Elapsed: 00:00:00.00
SQL> create index ind_acc_name on accanm17(name);
Index created.

Elapsed: 00:00:00.02
SQL> select * from accanm17;

ACC_ID TRANS_ID NAME BALANCE
--- --- --- ---
101 501 shradha 50000
102 502 rahul 40000
103 503 aditya 60000
Elapsed: 00:00:00.00
SQL> ^C
SQL> drop index ind_acc_name;
Index dropped.
Elapsed: 00:00:00.43
SQL> create view test_view as select
acc_id,cust_id,balance from accanm17 where balance >
30000; create view test_view as select
acc_id,cust_id,balance from accanm17 where balance
> 30000
*
ERROR at line 1:
ORA-00904: "CUST_ID": invalid identifier

Elapsed: 00:00:00.00
SQL> create view test_view as select
acc_id,trans_id,balance from accanm17 where balance >
30000;
View created.
Elapsed: 00:00:00.07
SQL> select * from test_view;
ACC_ID TRANS_ID BALANCE
--- --- ---
101 501 50000
102 502 40000

103 503 60000
Elapsed: 00:00:00.01
SQL> create view test_view as select
acc_id,trans_id,balance from accanm17 where balance >
4000; create view test_view as select
acc_id,trans_id,balance from accanm17 where balance
> 4000
*
ERROR at line 1:

ORA-00955: name is already used by an existing object

Elapsed: 00:00:00.00
SQL> create view check_view as select
acc_id,trans_id,balance from accanm17 where balance
> 30000;
View created.
Elapsed: 00:00:00.02
SQL> select * from check_view;
ACC_ID TRANS_ID BALANCE
--- --- ---
101 501 50000
102 502 40000
103 503 60000
Elapsed: 00:00:00.00
SQL> create view tick_view as select
acc_id,trans_id,balance from accanm17 where balance >
45000;
View created.
Elapsed: 00:00:00.02

SQL> select * from tick_view;
ACC_ID TRANS_ID BALANCE
---------- ---------- ----------
101 501 50000
103 503 60000
Elapsed: 00:00:00.01
SQL> alter table accanm17 add cust_id Number(5);
Table altered.
Elapsed: 00:00:00.08

4) Joins + Subqueries

SQL> create table stud17(sid number(5) primary key,sname

varchar2(10),dept_name varchar2(10),year number(5),total_marks

number(5));

Table created.

SQL> insert into stud17 values(&sid,'&sname','&dept_name',&year,&total_marks);

Enter value for sid: 101

Enter value for sname: Anmol

Enter value for dept_name: Comp

Enter value for year: 2

Enter value for total_marks: 597

old 1: insert into stud17 values(&sid,'&sname','&dept_name',&year,&total_marks)

new 1: insert into stud17 values(101,'Anmol','Comp',2,597)

1 row created.

SQL> /

Enter value for sid: 102

Enter value for sname: Arnav

Enter value for dept_name: Comp

Enter value for year: 2

Enter value for total_marks: 557

old 1: insert into stud17 values(&sid,'&sname','&dept_name',&year,&total_marks)

new 1: insert into stud17 values(102,'Arnav','Comp',2,557)

1 row created.

SQL> /

Enter value for sid: 103

Enter value for sname: Saksham

Enter value for dept_name: Ece

Enter value for year: 3

Enter value for total_marks: 499

old 1: insert into stud17 values(&sid,'&sname','&dept_name',&year,&total_marks)

new 1: insert into stud17 values(103,'Saksham','Ece',3,499)

1 row created.

SQL> /

Enter value for sid: 104

Enter value for sname: Shourya

Enter value for dept_name: It

Enter value for year: 3

Enter value for total_marks: 585

old 1: insert into stud17 values(&sid,'&sname','&dept_name',&year,&total_marks)

new 1: insert into stud17 values(104,'Shourya','It',3,585)

1 row created.

SQL> create table course18(cid number(5) primary key,cname

varchar2(10),credits number(5));

Table created.

SQL> insert into course18 values(&cid,'&cname',&credits);

Enter value for cid: 1

Enter value for cname: Dbms

Enter value for credits: 4

old 1: insert into course18 values(&cid,'&cname',&credits)

new 1: insert into course18 values(1,'Dbms',4)

1 row created.

SQL> /

Enter value for cid: 2

Enter value for cname: Oopcg

Enter value for credits: 3

old 1: insert into course18 values(&cid,'&cname',&credits)

new 1: insert into course18 values(2,'Oopcg',3)

1 row created.

SQL> /

Enter value for cid: 3

Enter value for cname: Pdt

Enter value for credits: 2

old 1: insert into course18 values(&cid,'&cname',&credits)

new 1: insert into course18 values(3,'Pdt',2)

1 row created.

SQL> /

Enter value for cid: 4

Enter value for cname: Dsa

Enter value for credits: 4

old 1: insert into course18 values(&cid,'&cname',&credits)

new 1: insert into course18 values(4,'Dsa',4)

1 row created.

SQL> create table enroll19(sid number(5),cid number(5),sem number(5),marks number(5),primary key(sid,cid),foreign key(sid) references stud17(sid),foreign key(cid) references course18(cid));

Table created.

SQL> insert into enroll19 values(&sid,&cid,&sem,&marks);

Enter value for sid: 101

Enter value for cid: 1

Enter value for sem: 2

Enter value for marks: 97

old 1: insert into enroll19 values(&sid,&cid,&sem,&marks)

new 1: insert into enroll19 values(101,1,2,97)

1 row created.

SQL> /

Enter value for sid: 102

Enter value for cid: 2

Enter value for sem: 1

Enter value for marks: 89

old 1: insert into enroll19 values(&sid,&cid,&sem,&marks)

new 1: insert into enroll19 values(102,2,1,89)

1 row created.

SQL> /

Enter value for sid: 103

Enter value for cid: 1

Enter value for sem: 3

Enter value for marks: 84

old 1: insert into enroll19 values(&sid,&cid,&sem,&marks)

new 1: insert into enroll19 values(103,1,3,84)

1 row created.

SQL> /

Enter value for sid: 102

Enter value for cid: 3

Enter value for sem: 4

Enter value for marks: 75

old 1: insert into enroll19 values(&sid,&cid,&sem,&marks)

new 1: insert into enroll19 values(102,3,4,75)

1 row created.

SQL> select * from stud17;

SID SNAME DEPT_NAME YEAR TOTAL_MARKS

--- --- --- --- ---

101 Anmol Comp 2 597

102 Arnav Comp 2 557

103 Saksham Ece 3 499

104 Shourya It 3 585

SQL> select * from course18;

CID CNAME CREDITS

--- --- ---

1 Dbms 4

2 Oopcg 3

3 Pdt 2

4 Dsa 4

SQL> select * from enroll19;

SID CID SEM MARKS

--- --- --- ---

101 1 2 97

103 2 1 89

103 1 3 84

102 3 4 75

SQL QUERIES

1) Display Students and Courses using INNER JOIN....!

SQL> select s.sname,c.cname from stud17 s join enroll19 e on s.sid=e.sid join course18 c on e.cid=c.cid;

SNAME CNAME

--- ---

Anmol Dbms

Arnav Oopcg

Arnav Pdt

Saksham Dbms

2) Display All Students and Courses using LEFT JOIN.....!

SQL> select s.sname,c.cname from stud17 s left join enroll19 e on

s.sid=e.sid left join course18 c on e.cid=c.cid;

SNAME CNAME

---------- ----------

Anmol Dbms

Saksham Dbms

Arnav Oopcg

Arnav Pdt

Shourya

3) Display All Courses and Students using RIGHT JOIN.....!

SQL> select s.sname,c.cname from stud17 s right join enroll19 e on

s.sid=e.sid right join course18 c on e.cid=c.cid;

SNAME CNAME

---------- ----------

Anmol Dbms

Arnav Oopcg

Arnav Pdt

Saksham Dbms

Dsa

4) Display Students from Same Department (SELF JOIN)....!

SQL> select s1.sname,s2.sname,s1.dept_name from stud17 s1 join

stud17 s2 on s1.dept_name=s2.dept_name;

SNAME SNAME DEPT_NAME

--- --- ---

Anmol Anmol Comp

Arnav Anmol Comp

Anmol Arnav Comp

Arnav Arnav Comp

Saksham Saksham Ece

Shourya Shourya It

OR

SQL> select s1.sname,s2.sname,s1.dept_name from stud17 s1 join stud17 s2 on

s1.dept_name=s2.dept_name and s1.sid<>s2.sid;

SNAME SNAME DEPT_NAME

---------- ---------- ----------

Arnav Anmol Comp

Anmol Arnav Comp

5) Display Students and Marks in a Particular Semester....!

SQL> select s.sname,e.marks from stud17 s join enroll19 e on s.sid=e.sid where e.sem=2;

SNAME MARKS

---------- ----------

Anmol 97

SUBQUERIES.....

1) Display Students Scoring More than Average Marks.....!

SQL> select sname from stud17 where total_marks>(select avg(total_marks) from stud17);

SNAME

Anmol

Shourya

2) Display Students Enrolled in DBMS.....!

SQL> select sname from stud17 where sid in(select sid from enroll19

where cid=(select cid from course18 where cname='Dbms'));

SNAME

Anmol

Saksham

3) Display Courses Having More Than One Student.....!

SQL> select c.cname, COUNT(e.sid) AS total_students FROM

course18 c JOIN enroll19 e ON c.cid = e.cid GROUP BY c.cname

HAVING COUNT(e.sid) > 1;

CNAME TOTAL_STUDENTS

--- ---

Dbms 2

4) Display Average Marks of Students in Each Course....!

SQL> select c.cname, AVG(e.marks) AS avg_marks FROM

course18 c JOIN enroll19 e ON c.cid = e.cid GROUP BY c.cname;

CNAME AVG_MARKS

---------- ----------

Dbms 90.5

Oopcg 89

Pdt 75

5) Procedure/Function

```
SQL> create table employee38(emp_id
number(5),emp_name
varchar2(10),emp_dept varchar2(10),emp_salary
number(5));
Table created.
SQL> insert into employee38
values(201,'Anmol','Comp',10000);
1 row created.
SQL> insert into employee38 values(202,'Jyoti','Mech',9000);
1 row created.
SQL> insert into employee38 values(203,'Aditya','It',8000);
1 row created.
SQL> select * from employee38;
EMP_ID EMP_NAME EMP_DEPT EMP_SALARY
---------- ---------- ---------- ----------
201 Anmol Comp 10000
202 Jyoti Mech 9000
203 Aditya It 8000
```

Procedure:

```
CREATE OR REPLACE PROCEDURE increase_salary
IS
BEGIN
-- Increase salary by 10%
UPDATE EMPLOYEE38
SET emp_salary = emp_salary + (emp_salary * 0.10);
COMMIT;
-- Display updated salaries
FOR rec IN (SELECT emp_id, emp_name, emp_salary FROM
EMPLOYEE38)
LOOP
DBMS_OUTPUT.PUT_LINE(
'ID: ' || rec.emp_id ||
' Name: ' || rec.emp_name ||
' Updated Salary: ' || rec.emp_salary
);
END LOOP;
END;
/

SQL> @/home/comp-pl-18/Documents/emp38_proc.sql;
Procedure created.
SQL> set serveroutput on;
SQL> exec increase_salary;
ID: 201 Name: Anmol Updated Salary: 11000
ID: 202 Name: Jyoti Updated Salary: 9900
ID: 203 Name: Aditya Updated Salary: 8800
PL/SQL procedure successfully completed.
SQL> 1`
```

Function:

```
CREATE OR REPLACE FUNCTION emp_avg_salary
RETURN NUMBER
IS
v_avg_salary NUMBER;
BEGIN
-- Calculate average salary
SELECT AVG(emp_salary)
INTO v_avg_salary
FROM EMPLOYEE38;
DBMS_OUTPUT.PUT_LINE('Average Salary: ' || v_avg_salary);
DBMS_OUTPUT.PUT_LINE('Employees having salary greater
than average:');
-- Display employees with salary greater than average
FOR rec IN (
SELECT emp_id, emp_name, emp_dept, emp_salary
FROM EMPLOYEE38
WHERE emp_salary > v_avg_salary
)
LOOP
DBMS_OUTPUT.PUT_LINE(
'ID: ' || rec.emp_id ||
' Name: ' || rec.emp_name ||
' Dept: ' || rec.emp_dept ||
' Salary: ' || rec.emp_salary
);
END LOOP;
RETURN v_avg_salary;
END;
/
```

```
SQL> @/home/comp-pl-
18/Documents/emp38_fun_avgsal.sql;
Function created.
SQL> set serveroutput on;
SQL> select emp_avg_salary from dual;
EMP_AVG_SALARY
--------------
9000
Average Salary: 9000
Employees having salary greater than average:
ID: 201 Name: Anmol Dept: Comp Salary: 10000
```

6) Triggers

```
SQL> create user anmol17 identified by mahajan;
User created.
SQL> grant connect,resource to anmol17;
Grant succeeded.
SQL> alter user anmol17 quota unlimited on users;
User altered.
SQL> alter user anmol17 account unlock;
User altered.
SQL> connect anmol17/mahajan; Connected.
SQL> create table emp17(emp_id number, emp_name
varchar(10), dept_id
number,emp_salary number);
Table created.
SQL> insert into emp17
values(&emp_id,'&emp_name',&dept_id,&emp_salary);
```

```
Enter value for emp_id: 101
Enter value for emp_name: Anmol
Enter value for dept_id: 1 Enter
value for emp_salary: 70000
old 1: insert into emp17
values(&emp_id,'&emp_name',&dept_id,&emp_salary) new
1: insert into emp17 values(101,'Anmol',1,70000)
1 row created.
SQL> /
Enter value for emp_id: 102
Enter value for emp_name: Saksham

Enter value for dept_id: 2 Enter
value for emp_salary: 60000
old 1: insert into emp17
values(&emp_id,'&emp_name',&dept_id,&emp_salary) new
1: insert into emp17 values(102,'Saksham',2,60000)
1 row created.
SQL> /
Enter value for emp_id: 103
Enter value for emp_name: Arnav
Enter value for dept_id: 1 Enter
value for emp_salary: 80000
old 1: insert into emp17
values(&emp_id,'&emp_name',&dept_id,&emp_salary) new
1: insert into emp17 values(103,'Arnav',1,80000)
1 row created.
SQL> /
Enter value for emp_id: 104
Enter value for emp_name: Shourya
```

```
Enter value for dept_id: 3 Enter
value for emp_salary: 50000
old 1: insert into emp17
values(&emp_id,'&emp_name',&dept_id,&emp_salary) new
1: insert into emp17 values(104,'Shourya',3,50000)
1 row created.
SQL> /
Enter value for emp_id: 105
Enter value for emp_name: Daredevil
Enter value for dept_id: 2 Enter
value for emp_salary: 67000
old 1: insert into emp17
values(&emp_id,'&emp_name',&dept_id,&emp_salary) new
1: insert into emp17 values(105,'Daredevil',2,67000)
1 row created.
SQL> select * from emp17;
EMP_ID EMP_NAME DEPT_ID EMP_SALARY
---------- ---------- ---------- ----------
101 Anmol 1 70000
102 Saksham 2 60000
103 Arnav 1 80000

104 Shourya 3 50000
105 Daredevil 2 67000
SQL> create table sal_audit17(audit_id number, emp_id
number, old_salary
number,new_salary number, change_date date, operation
varchar(15));
Table created.
```

```
SQL> create sequence audit_id_seq start with 1 increment
by 1;
Sequence created.

SQL> @/home/comp-pl-18/triggers17.sql;
Trigger created.

SQL> update emp17 set emp_salary = 70000 where
emp_id=105;
1 row updated.
SQL> update emp17 set emp_salary = 80000 where
emp_id=101;
1 row updated.
SQL> update emp17 set emp_salary = 85000where
emp_name = 'Shourya';
1 row updated.
SQL> SELECT * FROM sal_audit17;
AUDIT_ID EMP_ID OLD_SALARY NEW_SALARY CHANGE_D
OPERATION
---------- ---------- ---------- ---------- -------- ---------------
1 105 67000 70000 07-04-26 UPDATE
2 101 70000 80000 07-04-26 UPDATE
3 104 50000 85000 07-04-26 UPDATE
```

TRIGGER :

```
CREATE OR REPLACE TRIGGER tri_emp01
AFTER UPDATE OF emp_salary ON emp17
FOR EACH ROW
BEGIN
INSERT INTO sal_audit17
```

```
(audit_id ,emp_id,old_salary ,new_salary
,change_date,operation)
VALUES
(audit_id_seq.NEXTVAL, :OLD.emp_id , :OLD.emp_salary
,:NEW.emp_salary,
SYSDATE , 'UPDATE');
END;
/
```

7) Cursors

```
SQL> create table old_num17(roll_no number primary
key,name varchar(10));

Table created.

SQL> truncate table old_num17;

Table truncated.

SQL> insert into old_num17 values(&roll_no,'&name');
Enter value for roll_no: 1
Enter value for name: Anmol
old 1: insert into old_num17 values(&roll_no,'&name')
new 1: insert into old_num17 values(1,'Anmol')

1 row created.

SQL> /
Enter value for roll_no: 2
```

```
Enter value for name: Arnav
old 1: insert into old_num17 values(&roll_no,'&name')

new 1: insert into old_num17 values(2,'Arnav')

1 row created.

SQL> /
Enter value for roll_no: 3
Enter value for name: Saksham
old 1: insert into old_num17 values(&roll_no,'&name')
new 1: insert into old_num17 values(3,'Saksham')

1 row created.

SQL> /
Enter value for roll_no: 4
Enter value for name: Anju
old 1: insert into old_num17 values(&roll_no,'&name')
new 1: insert into old_num17 values(4,'Anju')

1 row created.

SQL> create table new_num17(roll_no number primary
key,name varchar(10));

Table created.

SQL> insert into new_num17 values(&roll_no,'&name');
Enter value for roll_no: 5
```

```
Enter value for name: Aarohi
old 1: insert into new_num17 values(&roll_no,'&name')

new 1: insert into new_num17 values(5,'Aarohi')

1 row created.

SQL> /
Enter value for roll_no: 6
Enter value for name: Mansi
old 1: insert into new_num17 values(&roll_no,'&name')
new 1: insert into new_num17 values(6,'Mansi')

1 row created.

SQL> select * from old_num17;

ROLL_NO NAME
---------- ----------
1 Anmol
2 Arnav
3 Saksham
4 Anju

SQL> select * from new_num17;

ROLL_NO NAME
---------- ----------
5 Aarohi
6 Mansi
```

```
SQL> SET SERVEROUTPUT ON;

CURSOR
DECLARE
CURSOR new_cursor IS
SELECT roll_no, name FROM new_num17;

CURSOR check_cursor(p_roll NUMBER) IS
SELECT roll_no
FROM old_num17
WHERE roll_no = p_roll;

v_roll old_num17.roll_no%TYPE;

BEGIN
FOR rec IN new_cursor LOOP
OPEN check_cursor(rec.roll_no);
FETCH check_cursor INTO v_roll;

IF check_cursor%NOTFOUND THEN
INSERT INTO old_num17 (roll_no, name)
VALUES (rec.roll_no, rec.name);
END IF;
CLOSE check_cursor;
END LOOP;
COMMIT;
DBMS_OUTPUT.PUT_LINE('Data merged successfully');
END;

/PL/SQL procedure successfully completed.
```

```
SQL> select * from new_num17;

ROLL_NO NAME
---------- ----------
5 Aarohi
6 Mansi

SQL> select * from old_num17;

ROLL_NO NAME
---------- ----------
1 Anmol
2 Arnav
3 Saksham
4 Anju
5 Aarohi
6 Mansi

6 rows selected.
```

8) MongoDB CRUD + Index

```
test> use companyDB switched
to db companyDB
companyDB> db.employee.insertMany([
... {
... emp_id: 101 ,
... name:"Anmol" ,
... qualification:"B.Tech" ,
```

```
... experience:3 ,
... salary: 45000 ,
... designation:"Developer"
... } ,
... {
... emp_id: 102 ,
... name:"Arnav" ,
... qualification:"B.Com" ,
... experience:2 ,
... salary: 40000 ,
... designation:"Manager"
... } ,
... {
... emp_id: 103 ,
... name:"Aditya" ,
... qualification:"B.Tech" ,
... experience:4 ,
... salary: 50000 ,
... designation:"Developer"
... }
... ])
{
acknowledged: true,
insertedIds: {
'0': ObjectId('69d388b4782b38508dc73bf8'),
'1': ObjectId('69d388b4782b38508dc73bf9'),
'2': ObjectId('69d388b4782b38508dc73bfa')
}
}
companyDB> db.employee.find()
```

[
{
_id: ObjectId('69d388b4782b38508dc73bf8'), emp_id:
101,
name: 'Anmol',
qualification: 'B.Tech',
experience: 3, salary:
45000,
designation: 'Developer'
},
{
_id: ObjectId('69d388b4782b38508dc73bf9'),
emp_id: 102,
name: 'Arnav',
qualification: 'B.Com',
experience: 2, salary:
40000,
designation: 'Manager'
},
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103,
name: 'Aditya',
qualification: 'B.Tech',
experience: 4, salary:
50000,
designation: 'Developer'
}
]

```
companyDB> show collections
employee employees
companyDB>
companyDB>
companyDB> db.employees.drop() true
companyDB> show collections employee
companyDB> db.employee.find({salary: {$gt: 40000} })
[
{
_id: ObjectId('69d388b4782b38508dc73bf8'),
emp_id: 101,
name: 'Anmol',
qualification: 'B.Tech',

experience: 3,
salary: 45000,
designation: 'Developer'
},
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103, name:
'Aditya', qualification:
'B.Tech', experience: 4,
salary: 50000,
designation: 'Developer'
}
]
companyDB> db.employee.find({ experience: {$gte: 3}})
[
{
```

```
_id: ObjectId('69d388b4782b38508dc73bf8'),
emp_id: 101,
name: 'Anmol',
qualification: 'B.Tech',
experience: 3, salary:
45000,
designation: 'Developer'
},
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103,
name: 'Aditya',
qualification: 'B.Tech',
experience: 4, salary:
50000,
designation: 'Developer'
}
]
companyDB> db.employee.find({designation: "Developer"})
[
{
_id: ObjectId('69d388b4782b38508dc73bf8'),
emp_id: 101,
name: 'Anmol',
qualification: 'B.Tech',
experience: 3,
salary: 45000,
designation: 'Developer'

},
```

```
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103,
name: 'Aditya',
qualification: 'B.Tech',
experience: 4, salary:
50000,
designation: 'Developer'
}
]
companyDB> db.employee.find({
... $and: [
... {designation: "Developer"} ,
... {salary : {$gt:40000} }
... ]
... })
[
{
_id: ObjectId('69d388b4782b38508dc73bf8'),
emp_id: 101,
name: 'Anmol',
qualification: 'B.Tech',
experience: 3, salary:
45000,
designation: 'Developer'
},
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103,
name: 'Aditya',
```

```
qualification: 'B.Tech',
experience: 4, salary:
50000,
designation: 'Developer'
}
]
companyDB> db.employee.find({
... $or: [
... {designation: "Developer"} ,
... {salary : {$gt:45000}}
... ]
... })
[

{
_id: ObjectId('69d388b4782b38508dc73bf8'),
emp_id: 101,
name: 'Anmol',
qualification: 'B.Tech',
experience: 3, salary:
45000,
designation: 'Developer'
},
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103,
name: 'Aditya',
qualification: 'B.Tech',
experience: 4, salary:
50000,
```

```
designation: 'Developer'
}
]
companyDB> db.employee.find({
... designation: {$ne: "Developer"}
... })
[
{
_id: ObjectId('69d388b4782b38508dc73bf9'),
emp_id: 102,
name: 'Arnav',
qualification: 'B.Com',
experience: 2, salary:
40000,
designation: 'Manager'
}
]
companyDB> show collections employee
companyDB> db.employee.find()
[
{
_id: ObjectId('69d388b4782b38508dc73bf8'),
emp_id: 101,
name: 'Anmol',
qualification: 'B.Tech',
experience: 3, salary:
45000,
designation: 'Developer'

},
```

```
{
_id: ObjectId('69d388b4782b38508dc73bf9'),
emp_id: 102,
name: 'Arnav',
qualification: 'B.Com',
experience: 2, salary:
40000,
designation: 'Manager'
},
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103, name:
'Aditya', qualification:
'B.Tech', experience: 4,
salary: 50000,
designation: 'Developer'
}
]
companyDB> db.employee.updateOne(
... {name:"Arnav"},
... {$set:{salary:42000}}
... )
{
acknowledged: true,
insertedId: null,
matchedCount: 1,
modifiedCount: 1,
upsertedCount: 0
}
```

```
companyDB> db.employee.updateMany( {},
{$inc:{salary:5000}} )
{
acknowledged: true,
insertedId: null,
matchedCount: 3,
modifiedCount: 3,
upsertedCount: 0
}
companyDB> db.employee.find()
[
{
_id: ObjectId('69d388b4782b38508dc73bf8'),
emp_id: 101,
name: 'Anmol',

qualification: 'B.Tech',
experience: 3, salary:
50000,
designation: 'Developer'
},
{
_id: ObjectId('69d388b4782b38508dc73bf9'),
emp_id: 102,
name: 'Arnav',
qualification: 'B.Com',
experience: 2, salary:
47000,
designation: 'Manager'
},
```

```
{
_id: ObjectId('69d388b4782b38508dc73bfa'),
emp_id: 103,
name: 'Aditya',
qualification: 'B.Tech',
experience: 4, salary:
55000,
designation: 'Developer'
}
]
companyDB>
```

9) MongoDB Aggregation

```
test> use companyDB switched to
db companyDB companyDB>
db.Employees.find()
[
{ _id: 1, name: 'Anmol Mahajan', dept: 'Comp', Marks: 90 },
{ _id: 2, name: 'Aditya Jagtap', dept: 'Comp', Marks: 95 },
{ _id: 3, name: 'Sujal Khobare', dept: 'IT', Marks: 92 },
{ _id: 4, name: 'Anjali Jagtap', dept: 'ECE', Marks: 89 },
{ _id: 5, name: 'Jyotiraditya Kale', dept: 'Mech', Marks: 69 }
]
companyDB> db.Employees.drop()
true companyDB> show
collections

companyDB> db.createCollection("employees")
{ ok: 1 }
```

```
companyDB> db.employees.insertMany([
... { emp_id: 1, name: "Anmol", dept: "IT", designation:
"Manager", salary: 80000 },
... { emp_id: 2, name: "Anjali ", dept: "HR", designation:
"Executive", salary: 40000 },
... { emp_id: 3, name: "Aditya ", dept: "IT", designation:
"Developer", salary: 65000 },
... { emp_id: 4, name: "Vansh", dept: "Finance", designation:
"Analyst", salary: 50000 },
... { emp_id: 5, name: "Sujal", dept: "HR", designation:
"Executive", salary: 40000 },
... { emp_id: 6, name: "Jyotiraditya", dept: "IT", designation:
"Manager", salary: 90000 }
... ])
{
acknowledged: true,
insertedIds: {
'0': ObjectId('69dcbc5565b10d3b9f86b01d'),

'1': ObjectId('69dcbc5565b10d3b9f86b01e'),
'2': ObjectId('69dcbc5565b10d3b9f86b01f'),
'3': ObjectId('69dcbc5565b10d3b9f86b020'),
'4': ObjectId('69dcbc5565b10d3b9f86b021'),
'5': ObjectId('69dcbc5565b10d3b9f86b022')
}
}
companyDB> db.employees.find()
[
{
_id: ObjectId('69dcbc5565b10d3b9f86b01d'),
```

emp_id: 1, name: 'Anmol',
dept: 'IT', designation:
'Manager', salary:
80000
},
{
_id: ObjectId('69dcbc5565b10d3b9f86b01e'),
emp_id: 2, name: 'Anjali ', dept: 'HR',
designation: 'Executive', salary: 40000
},
{
_id: ObjectId('69dcbc5565b10d3b9f86b01f'),
emp_id: 3, name: 'Aditya ', dept: 'IT',
designation: 'Developer',
salary: 65000
},
{
_id: ObjectId('69dcbc5565b10d3b9f86b020'),
emp_id: 4, name: 'Vansh', dept: 'Finance',
designation: 'Analyst', salary: 50000

},
{
_id: ObjectId('69dcbc5565b10d3b9f86b021'),
emp_id: 5, name: 'Sujal', dept: 'HR',
designation: 'Executive', salary: 40000
},
{
_id: ObjectId('69dcbc5565b10d3b9f86b022'),
emp_id: 6, name: 'Jyotiraditya', dept: 'IT',

```
designation: 'Manager', salary: 90000
}
]
companyDB> db.employees.aggregate([
... {
... $group: {
... _id: "$dept",
... avgSalary: { $avg: "$salary" }
... }
... }
... ])
[
{ _id: 'Finance', avgSalary: 50000 },
{ _id: 'IT', avgSalary: 78333.33333333333 },
{ _id: 'HR', avgSalary: 40000 }
]
companyDB> db.employees.aggregate([
... {
... $group: {
... _id: "$designation",
... count: { $sum: 1 }
... }

... }
... ])
[
{ _id: 'Executive', count: 2 },
{ _id: 'Manager', count: 2 },
{ _id: 'Developer', count: 1 },
{ _id: 'Analyst', count: 1 }
```

```
]
companyDB> db.employees.aggregate([
... {
... $sort: { salary: -1 }
... }
... ])
[
{
_id: ObjectId('69dcbc5565b10d3b9f86b022'),
emp_id: 6, name: 'Jyotiraditya', dept: 'IT',
designation: 'Manager', salary: 90000
},
{
_id: ObjectId('69dcbc5565b10d3b9f86b01d'),
emp_id: 1, name: 'Anmol',
dept: 'IT', designation:
'Manager', salary:
80000
},
{
_id: ObjectId('69dcbc5565b10d3b9f86b01f'),
emp_id: 3, name: 'Aditya ', dept: 'IT',
designation: 'Developer', salary: 65000
},

{
_id: ObjectId('69dcbc5565b10d3b9f86b020'),
emp_id: 4, name: 'Vansh', dept: 'Finance',
designation: 'Analyst', salary: 50000
},
```

```
{
_id: ObjectId('69dcbc5565b10d3b9f86b01e'),
emp_id: 2, name: 'Anjali ', dept: 'HR',
designation: 'Executive', salary: 40000
},
{
_id: ObjectId('69dcbc5565b10d3b9f86b021'),
emp_id: 5, name: 'Sujal', dept: 'HR',
designation: 'Executive', salary: 40000
}
]
companyDB> db.employees.aggregate([
... {
... $match: { salary: { $gt: 60000 } }
... }
... ])
[
{
_id: ObjectId('69dcbc5565b10d3b9f86b01d'),
emp_id: 1, name: 'Anmol',
dept: 'IT', designation:
'Manager', salary:
80000
},
{

_id: ObjectId('69dcbc5565b10d3b9f86b01f'),
emp_id: 3, name: 'Aditya ', dept: 'IT',
designation: 'Developer', salary: 65000
},
```

```
{
_id: ObjectId('69dcbc5565b10d3b9f86b022'),
emp_id: 6, name: 'Jyotiraditya', dept: 'IT',
designation: 'Manager', salary: 90000
}
]
companyDB> db.employees.aggregate([
... {
... $match: { dept: "IT", salary: { $gt: 60000 } }
... },
... {
... $group: {
... _id: "$dept",
... avgSalary: { $avg: "$salary" }
... }
... }
... ])
[ { _id: 'IT', avgSalary: 78333.33333333333 } ]
companyDB>
```